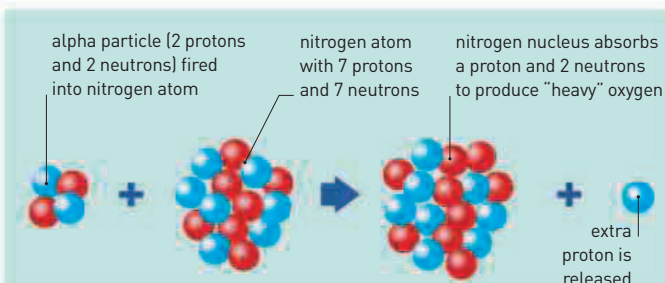


300 THOUSAND LIGHT YEARS

SHAPLEY'S ESTIMATE OF THE DIAMETER OF THE MILKY WAY

American astronomer Harlow Shapley's first measurement of the Milky Way galaxy was an overestimate. It is 100,000 light-years across and contains more than 100 billion stars.



THE FIRST TRANSMUTATION

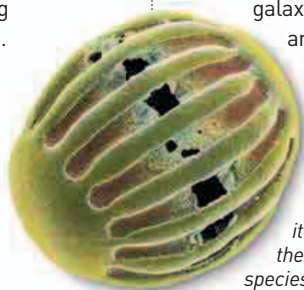
British physicist Ernest Rutherford was the first person to cause one element to change into another—a phenomenon called transmutation. He showed that it was possible to change the size of an atom's nucleus by bombarding it with other particles. Later scientists exploited this process to release huge amounts of energy.

IN 1916, AMERICAN CHEMIST GILBERT LEWIS (1875–1946) suggested that when atoms bonded together to make bigger molecules, they **shared their outer electrons**. In 1919, his colleague Irving Langmuir (1881–1957) would expand the idea and call them covalent bonds.

Swedish geologist Lennart von Post (1884–1951) explained a new way of studying geological deposits. Different **plant species produce distinctly shaped pollen grains**, and von Post was able to identify pollen to make conclusions about

the historic vegetation of peat beds. The study of pollen (palynology) is still an important analytical study used for example in forensics.

American astronomer Harlow Shapley (1885–1972) started a **photographic survey of star clusters**. His systematic measurements would show that clusters surround our galaxy like a halo, and are not centered on Earth.



Pollen grain
The distinctive sculpturing of pollen grains makes it possible to identify the source plant species by microscopy.

In 1917, at the Mount Wilson Observatory in California, the assembly of the **Hooker reflecting telescope**—the largest in the world at the time—was finished. To this day, its 100in (254 cm) mirror is the largest solid-glass mirror ever made. The telescope was used to make the first measurements of stars other than the Sun.

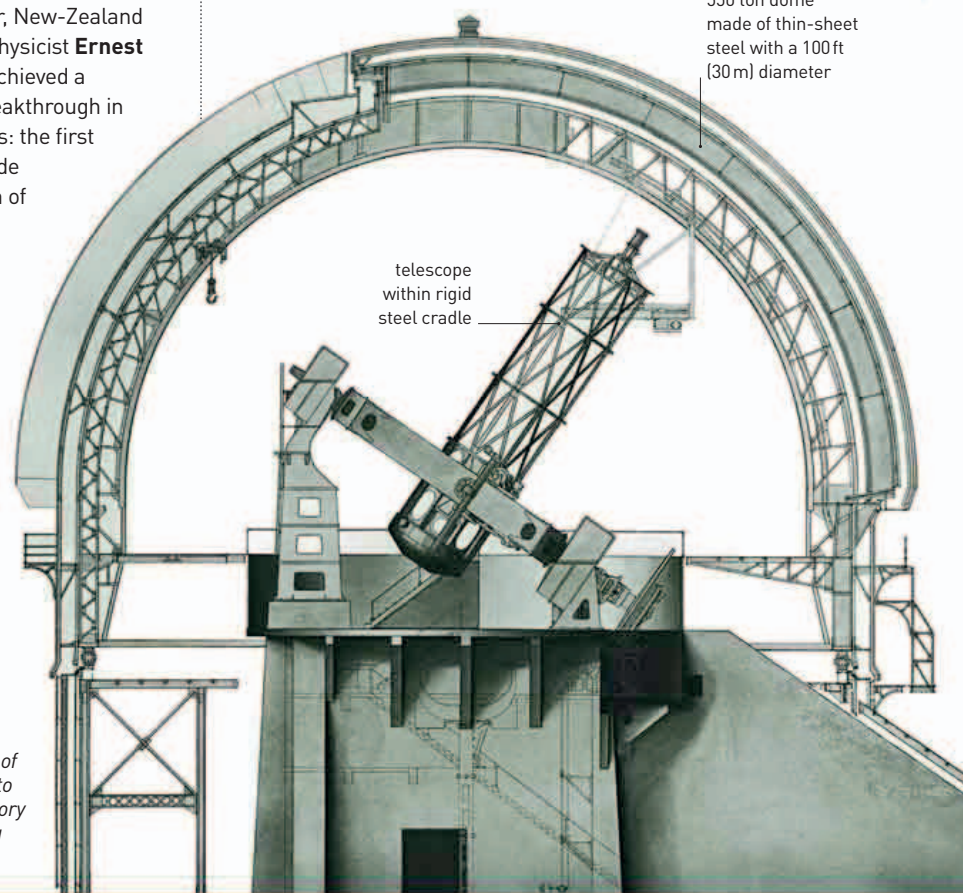
In November, New-Zealand born British physicist **Ernest Rutherford** achieved a significant breakthrough in atomic physics: the first ever man-made transmutation of elements. He succeeded in **turning nitrogen**

Hooker telescope
This telescope was used to make the first measurements of star sizes, and to develop the theory of an expanding universe.

atoms into a form of oxygen, by firing them with alpha particles.

When the US entered World War I in April 1917, British-born medical officer **Oswald Robertson** (1886–1966) introduced a **blood bank system** to the frontline. The Royal Army Medical Corps called it the most important medical advance of

the war. Treating blood with **anticoagulant made blood transfusions faster and safer**. British physician **Arthur Cushny** (1866–1926) published the **first major study of kidney physiology**. He correctly deduced that the organ filtered blood and reabsorbed nutrients to produced waste urine.



January 1, 1916 Royal Army Medical Corps performs first blood transfusion using stored and cooled blood

April 1916 Gilbert Lewis describes **chemical bonding** in terms of sharing outer electrons

July 21, 1916 Einstein describes a **theory of radiation emission**

July 10–15, 1916 Lennart von Post introduces **pollen analysis**

1916 Harlow Shapley starts to photograph star clusters, leading him to determine the **size and shape of the Milky Way**

1916 American ecologist Frederic Clements publishes **Plant Succession**

1917 French microbiologist Felix d'Hérelle describes viruses that can destroy bacteria: **bacteriophages**

1917 Arthur Cushny publishes *The Secretion of Urine*, describing how the **kidneys form urine by filtration**

mid-1917 French physicist Paul Langevin tests an **echo sounding system** for detecting submarines

October 1917 American zoologist Joseph Grinnell establishes the idea of an **ecological niche**

November 1, 1917 Hooker telescope at Mount Wilson Observatory, California, is used for the first time